UNIVERSITÄT BONN IGG

Application and Evaluation of Kinematic MSS

Evaluation of MMS

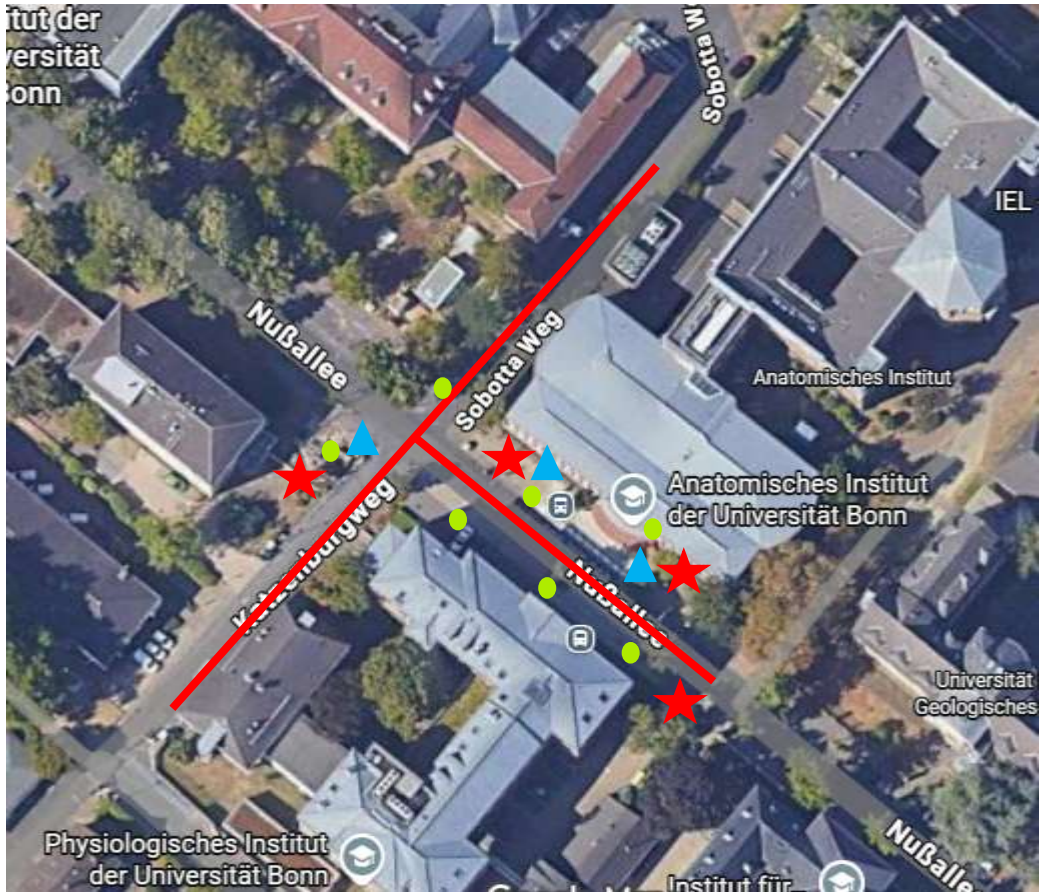
Team members: Abu Talha, Sajjad Hussain, Shuyi Wu, Moein Taherkhani

Outline

- Scanning Plan
- Static Part
- Kinematic Part
- Evaluation

Scanning Plan

Scanning Plan





★ **GNSS receiver**

— **Kinematic Measurement Trajectory**

● **Target**

▲ **TLS**

Static Part

Instrument introduction

Static measurements



Terrestrial Laser Scanner

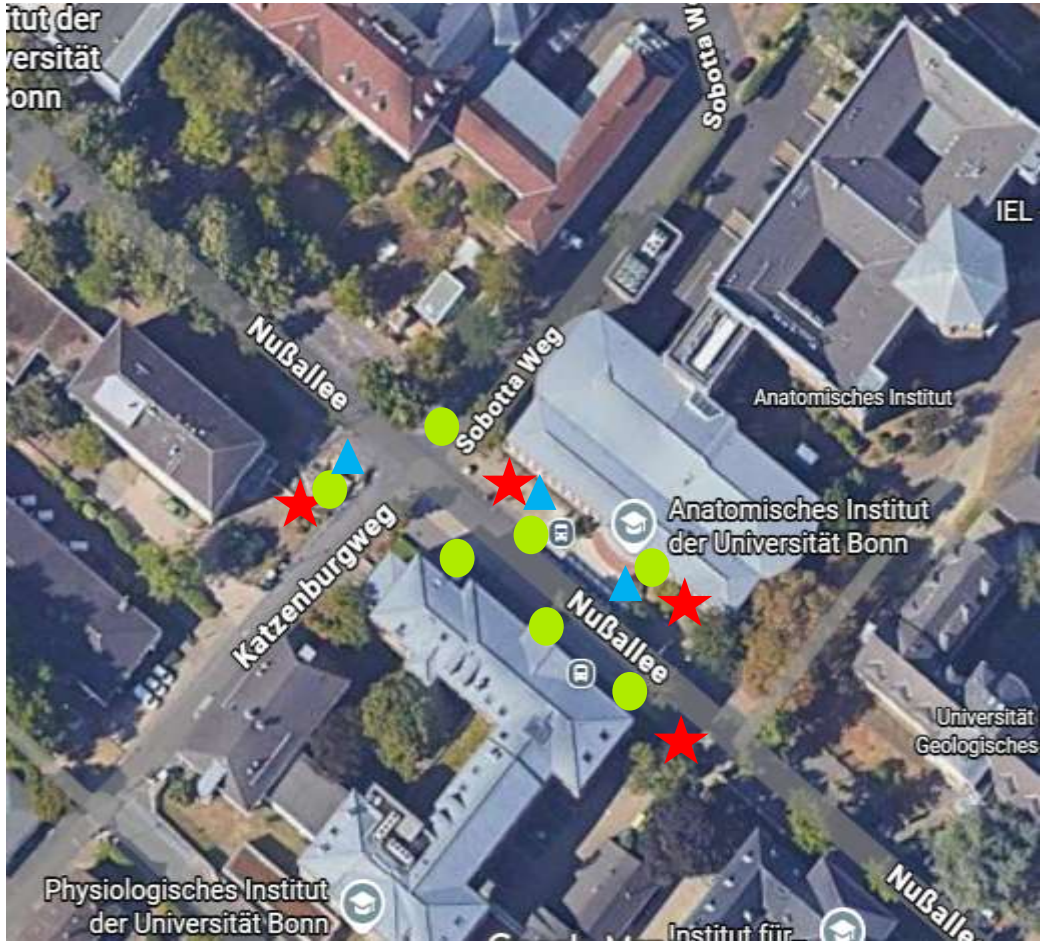


GNSS Receiver



Artificial Targets

Data Acquisition – Static







Set GNSS Receiver → **Registration**

Georeferencing – Static

Data Acquisition



GNSS



Target



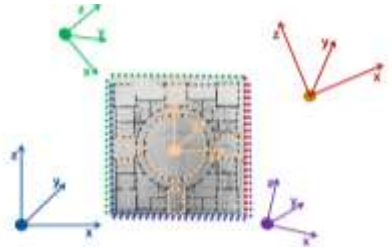
TLS

TLS Georeferenced

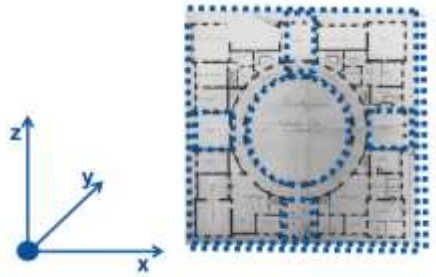


I

Registration



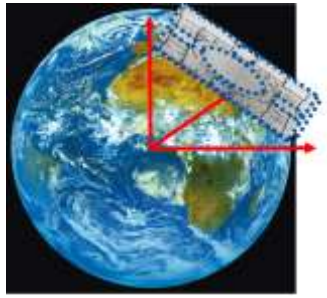
merging



II

Georeferencing

Local Coordinate System → Global Coordinate System (e.g. WGS84)



III

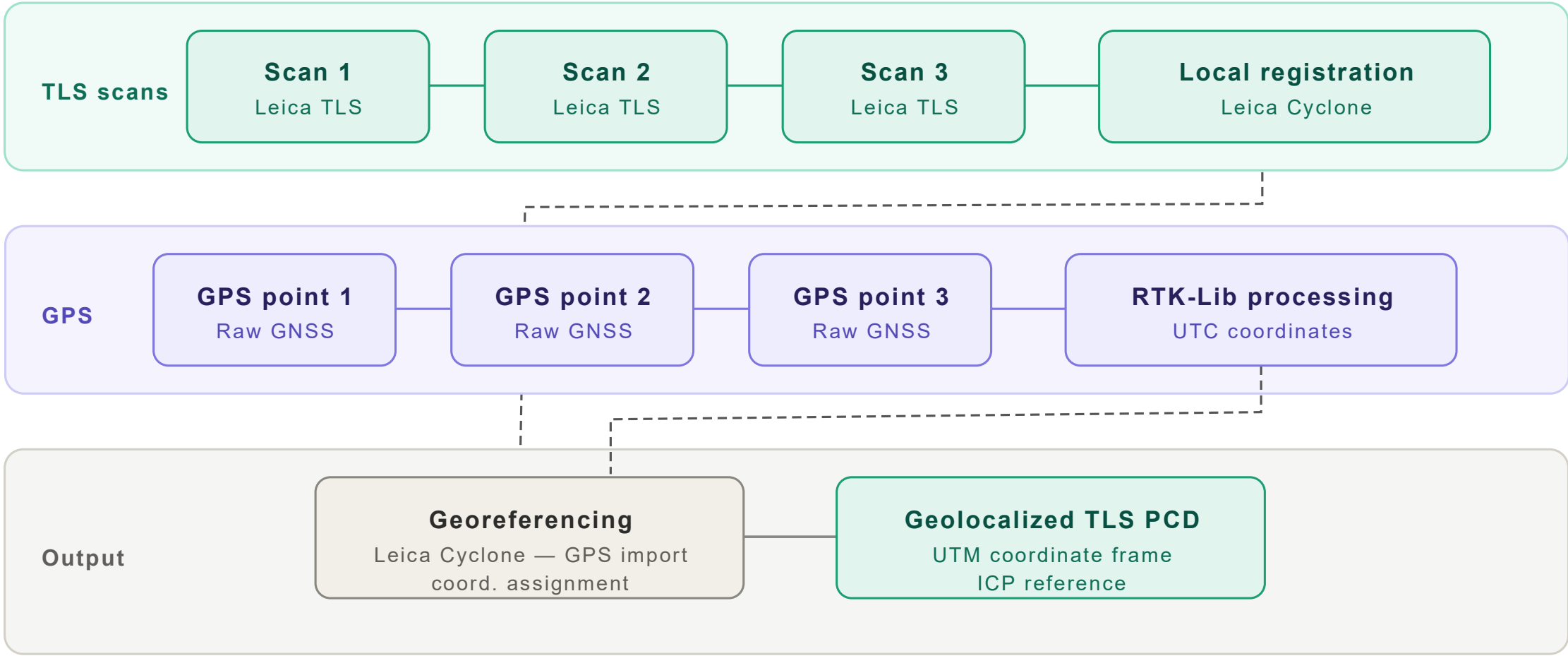
Projection (Optional)

Position + Height

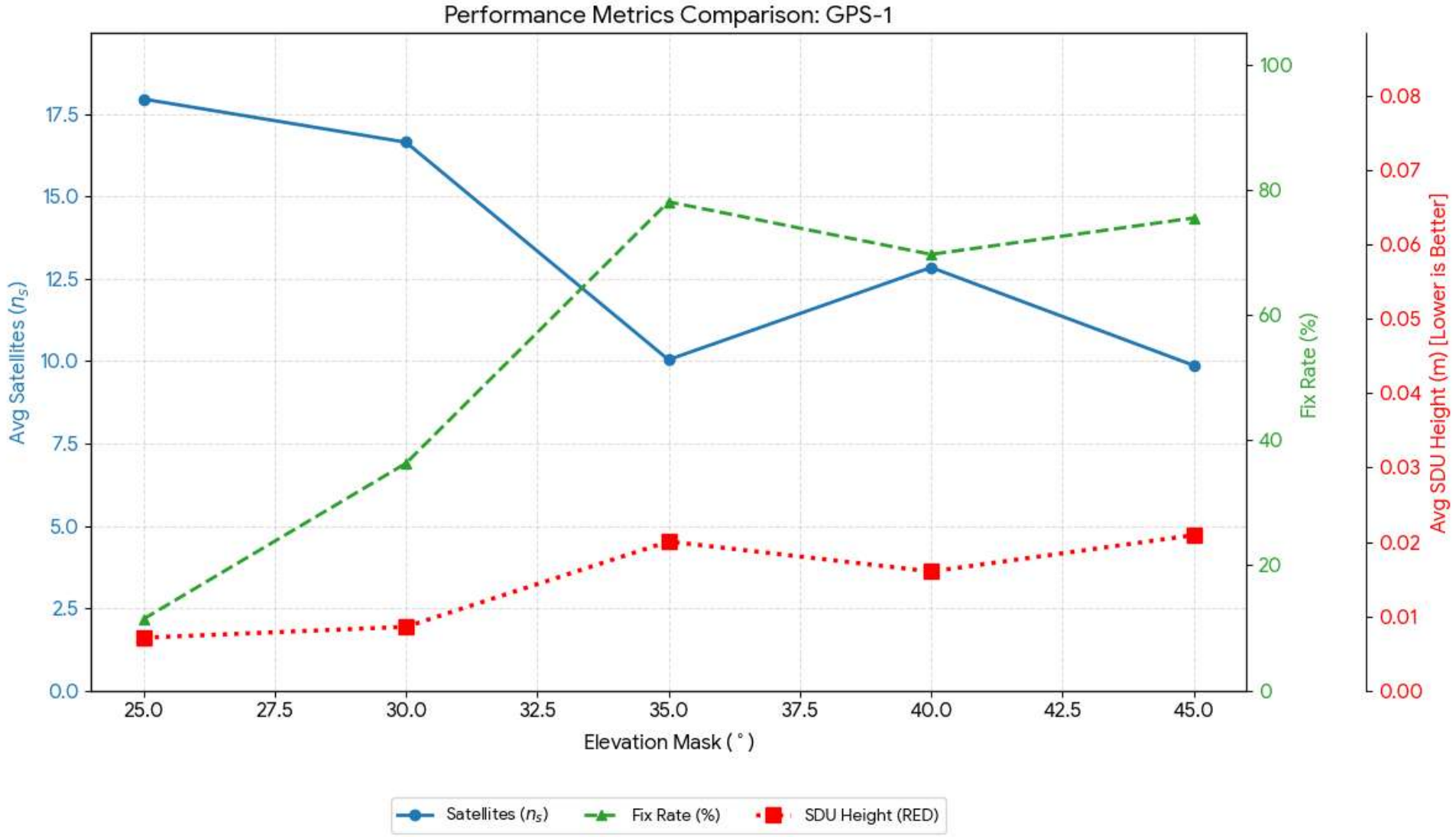
(2D) (1D)



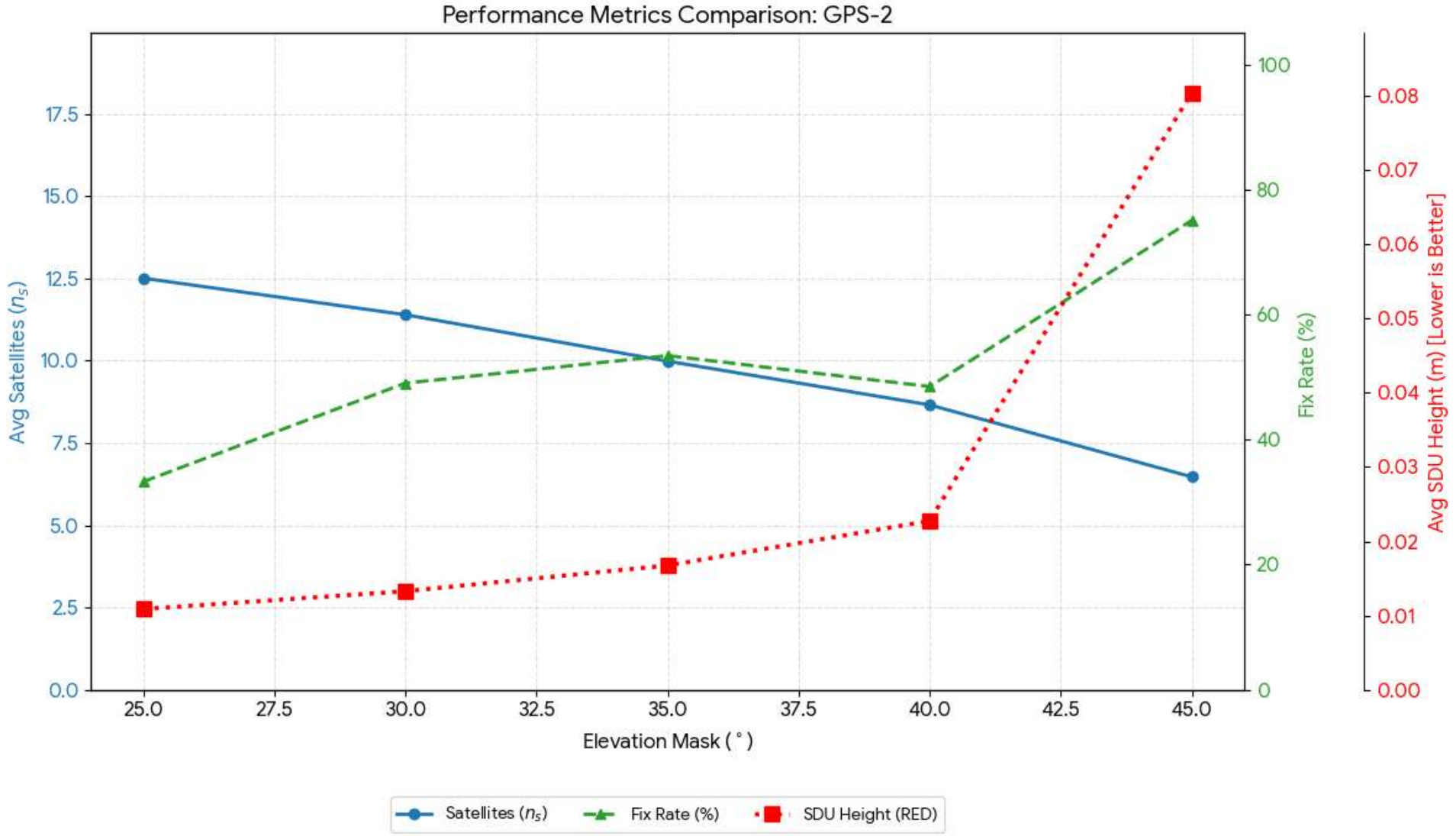
Georeferencing – Static



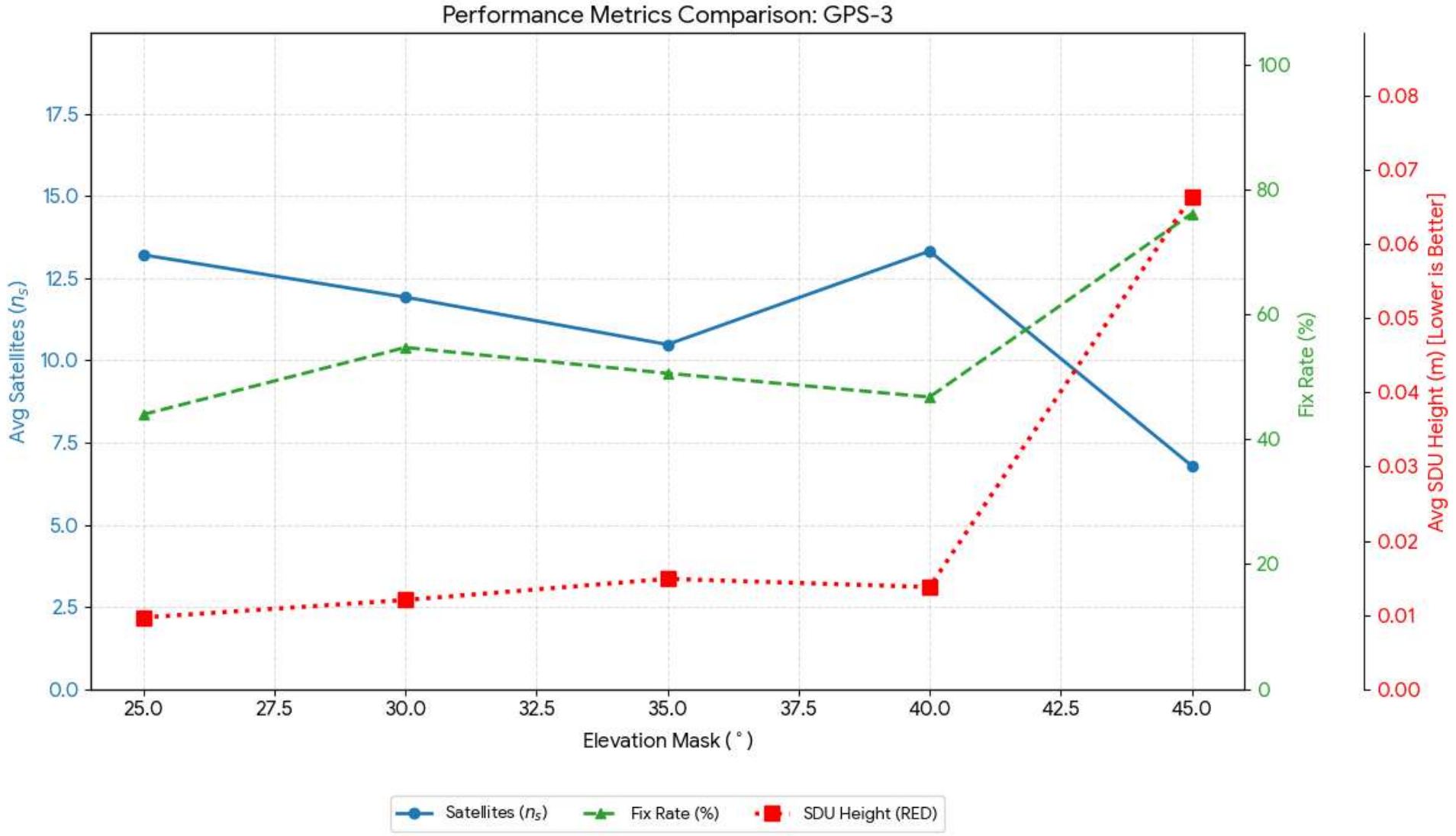
GNSS Processing – GPS 1



GNSS Processing – GPS 2



GNSS Processing – GPS 3



GNSS Processing - Summary

Configuration: 35.0° Elevation Mask | GPS + GLONASS | Dual-Freq Kinematic RTK

GPS-1		GPS-2		GPS-3	
STATIC SOLUTION		STATIC SOLUTION		STATIC SOLUTION	
Lat:	50.72646509°	Lat:	50.72667997°	Lat:	50.72683911°
Lon:	7.08927958°	Lon:	7.08912624°	Lon:	7.08878314°
Hgt:	109.712m	Hgt:	109.393m	Hgt:	109.131m
FIX RATE	78.1%	FIX RATE	53.4%	FIX RATE	50.6%
PRECISION (STD DEV)		PRECISION (STD DEV)		PRECISION (STD DEV)	
σ North:	0.0742 m	σ North:	0.0636 m	σ North:	0.0727 m
σ East:	0.0351 m	σ East:	0.0325 m	σ East:	0.0338 m
σ Vertical:	0.2945 m	σ Vertical:	0.3469 m	σ Vertical:	0.3089 m



Static PCD Registration

UNIVERSITÄT BONN IGG

Local Registration Errors

Constraint ID	ScanWorld	ScanWorld	Type	Status	Weight	Overlap Points	Error	Error Vector	Group Error	Group Error Vector	Group
g2	Station-002: S...	Station-005: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.002 m	(0.000, 0.002, 0.001) m	n/a	n/a	Ungrouped
g2	Station-002: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.005 m	(0.001, -0.001, 0.005) m	n/a	n/a	Ungrouped
T3	Station-002: S...	Station-005: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.002 m	(0.001, -0.001, -0.002) m	n/a	n/a	Ungrouped
T3	Station-002: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.008 m	(-0.004, 0.002, -0.007) m	n/a	n/a	Ungrouped
T4	Station-002: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.004 m	(0.002, 0.001, -0.003) m	n/a	n/a	Ungrouped
T1	Station-002: S...	Station-005: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.002 m	(-0.001, 0.000, 0.002) m	n/a	n/a	Ungrouped
T1	Station-002: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.003 m	(0.000, -0.001, 0.003) m	n/a	n/a	Ungrouped
T2	Station-002: S...	Station-005: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.001 m	(0.000, -0.001, -0.001) m	n/a	n/a	Ungrouped
T2	Station-002: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.002 m	(0.001, -0.001, 0.001) m	n/a	n/a	Ungrouped
T3	Station-005: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.008 m	(-0.005, 0.003, -0.005) m	n/a	n/a	Ungrouped
g2	Station-005: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.005 m	(0.001, -0.003, 0.004) m	n/a	n/a	Ungrouped
T1	Station-005: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.002 m	(0.001, -0.001, 0.002) m	n/a	n/a	Ungrouped
T2	Station-005: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.002 m	(0.002, 0.000, 0.002) m	n/a	n/a	Ungrouped
T5	Station-005: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.001 m	(0.000, 0.000, 0.001) m	n/a	n/a	Ungrouped
g4	Station-005: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.004 m	(0.001, 0.000, -0.004) m	n/a	n/a	Ungrouped
g3	Station-005: S...	Station-007: S...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.002 m	(0.002, 0.000, 0.001) m	n/a	n/a	Ungrouped

Georeferencing Errors

Constraint ID	ScanWorld	ScanWorld	Type	Status	Weight	Overlap Points	Error	Error Vector	Group Error	Group Error Vector	Group
g1	coordinates.txt...	ScanWorld [lo...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.006 m	(-0.001, -0.005, 0.001) m	n/a	n/a	Ungrouped
g2	coordinates.txt...	ScanWorld [lo...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.021 m	(0.004, 0.007, 0.019) m	n/a	n/a	Ungrouped
g3	coordinates.txt...	ScanWorld [lo...	Coincident: Vertex - Vertex	On	1.0000	n/a	0.020 m	(-0.002, -0.002, -0.020) m	n/a	n/a	Ungrouped

Georeferenced PCD – Static



TLS Georeferenced Pointcloud

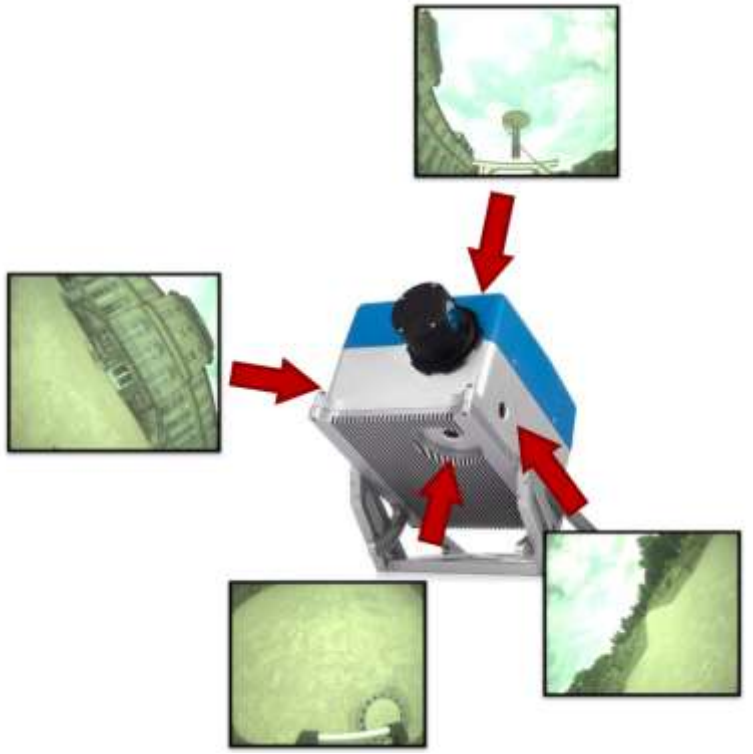
Kinematic Part

Instrument introduction

Kinematic measurements

Z+F Profiler 9020c

- **Scan rate:** 53 – 267 rps
  up to 2 MHz
- **Distance:**
  min. 0.3 m
  max. 182 m
- **Angular accuracy:** 0.007°
- **Distance measurement precision:**
  - Depends on distance, surface properties, configuration,…
- **Internal memory:** 256 GB



- 4x 5.05 MPx Fish eye cameras
  (top, left, right, bottom)
- Field of view: 120° x 6
- Fully calibrated syster
- Mainly for colorizatio

Data Acquisition – Kinematic







Kinematic Measurement Trajectory

- **Start Point**
- **End Point**

Pointcloud Registration - Kinematic

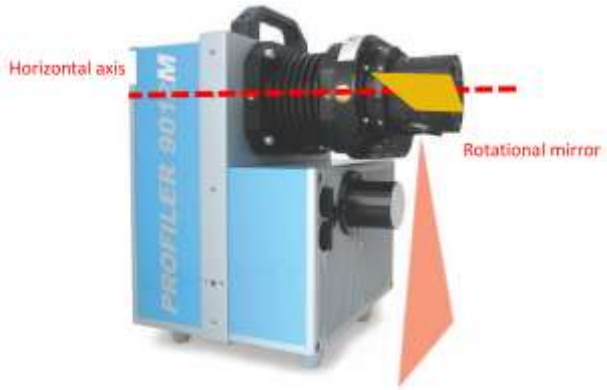
UNIVERSITÄT BONN IGG

Mobile Laser Scanning Transformation

$$\mathbf{p}_{object}^{global}(t_s) = \mathbf{T}_{body}^{global}(t_s) \cdot \mathbf{T}_{sensor}^{body} \cdot \mathbf{p}_{object}^{sensor}(t_s) \Rightarrow \begin{bmatrix} x_e \\ y_e \\ z_e \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L,B)\,\mathbf{R}_b^n(\phi,\theta,\psi) \cdot \left(\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + \mathbf{R}_s^b(\alpha,\beta,\gamma) \cdot \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix} \right)$$

Sensor Synchronisation

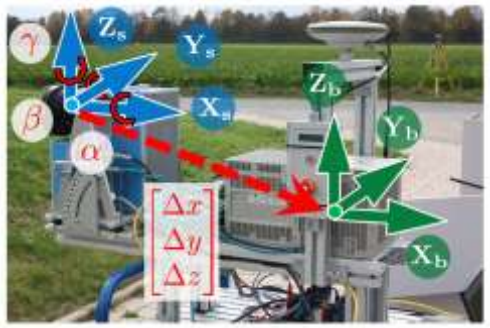
I. Sensor Frame Profile Scanner



$$\begin{bmatrix} x_s \\ y_s \\ z_s \end{bmatrix} = \begin{bmatrix} d \cdot \sin b \cdot \sin a \\ d \cdot \sin b \cdot \cos a \\ d \cdot \cos b \end{bmatrix} = \begin{bmatrix} 0 \\ d \cdot \sin b \\ d \cdot \cos b \end{bmatrix}$$

No rotation around the vertical axis (a=0)

II. System Calibration



Boresight angles

α, β, γ

Level arm

Δ x, Δ y, Δ z

$$\mathbf{R}_s^b(\alpha,\beta,\gamma) = \begin{bmatrix} \cos\beta \cdot \cos\gamma & \cos\alpha \cdot \sin\gamma + \cos\gamma \cdot \sin\alpha \cdot \sin\beta & \sin\alpha \cdot \sin\gamma - \cos\alpha \cdot \cos\gamma \cdot \sin\beta \\ -\cos\beta \cdot \sin\gamma & \cos\alpha \cdot \cos\gamma - \sin\alpha \cdot \sin\beta \cdot \sin\gamma & \cos\gamma \cdot \sin\alpha + \cos\alpha \cdot \sin\beta \cdot \sin\gamma \\ \sin\beta & -\cos\beta \cdot \sin\alpha & \cos\alpha \cdot \cos\beta \end{bmatrix}$$

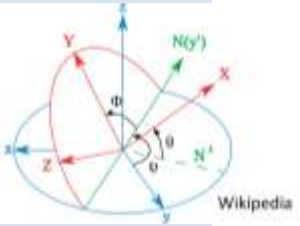
$$\begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} + \mathbf{R}_s^b(\alpha,\beta,\gamma)$$

III. Trajectory

Body Frame → Navigation Frame

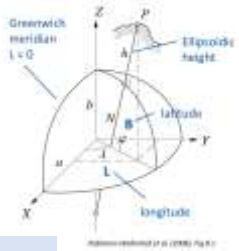
$$\mathbf{R}_b^n(\phi,\theta,\psi)$$

Roll Φ, Pitch θ, Yaw ψ



Navigation Frame→ Earth Frame

$$\begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \mathbf{R}_n^e(L,B)\,\mathbf{R}_b^n(\phi,\theta,\psi)$$



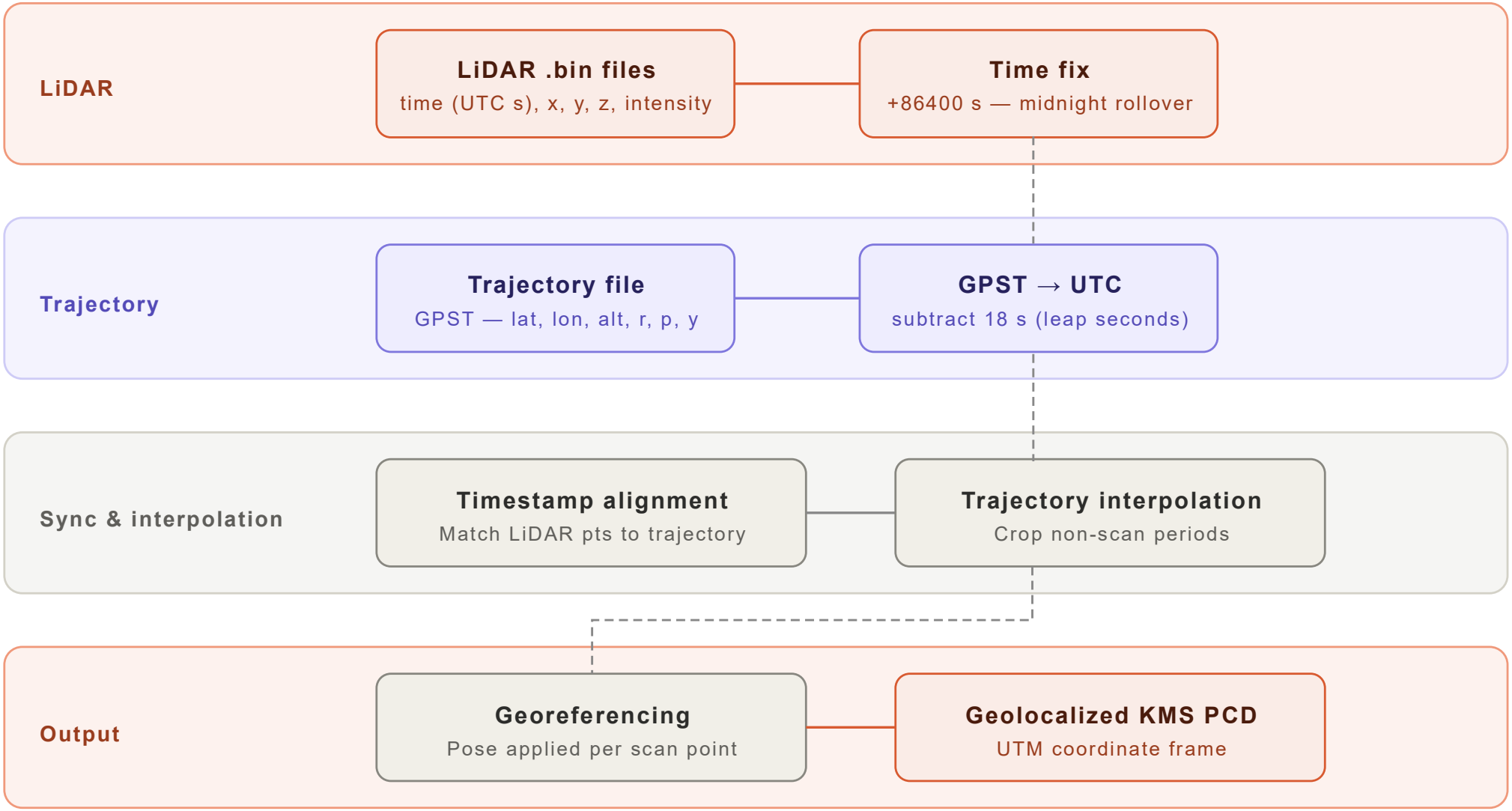
Projection (Optional)

Position + Height

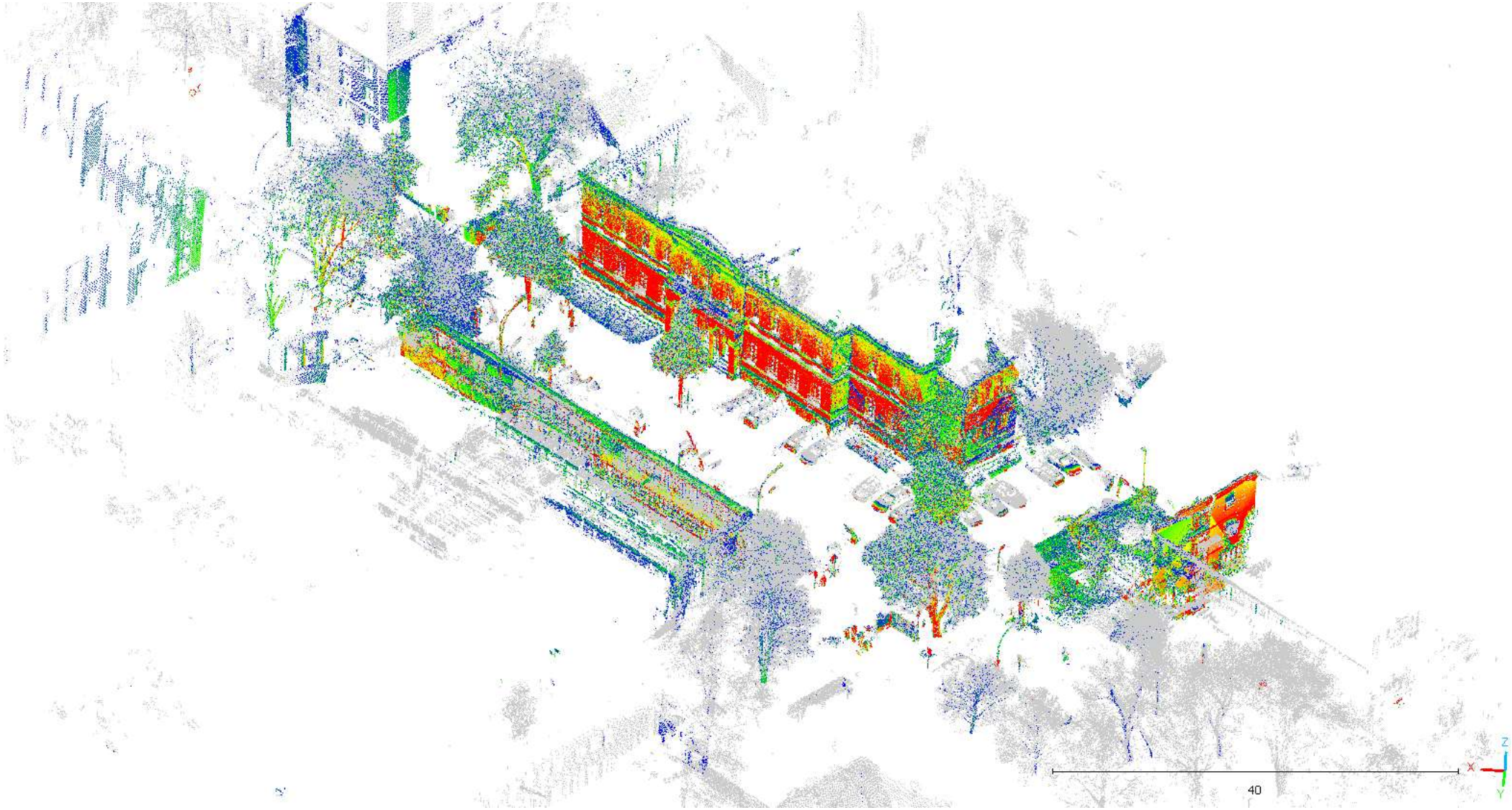
(2D) (1D)



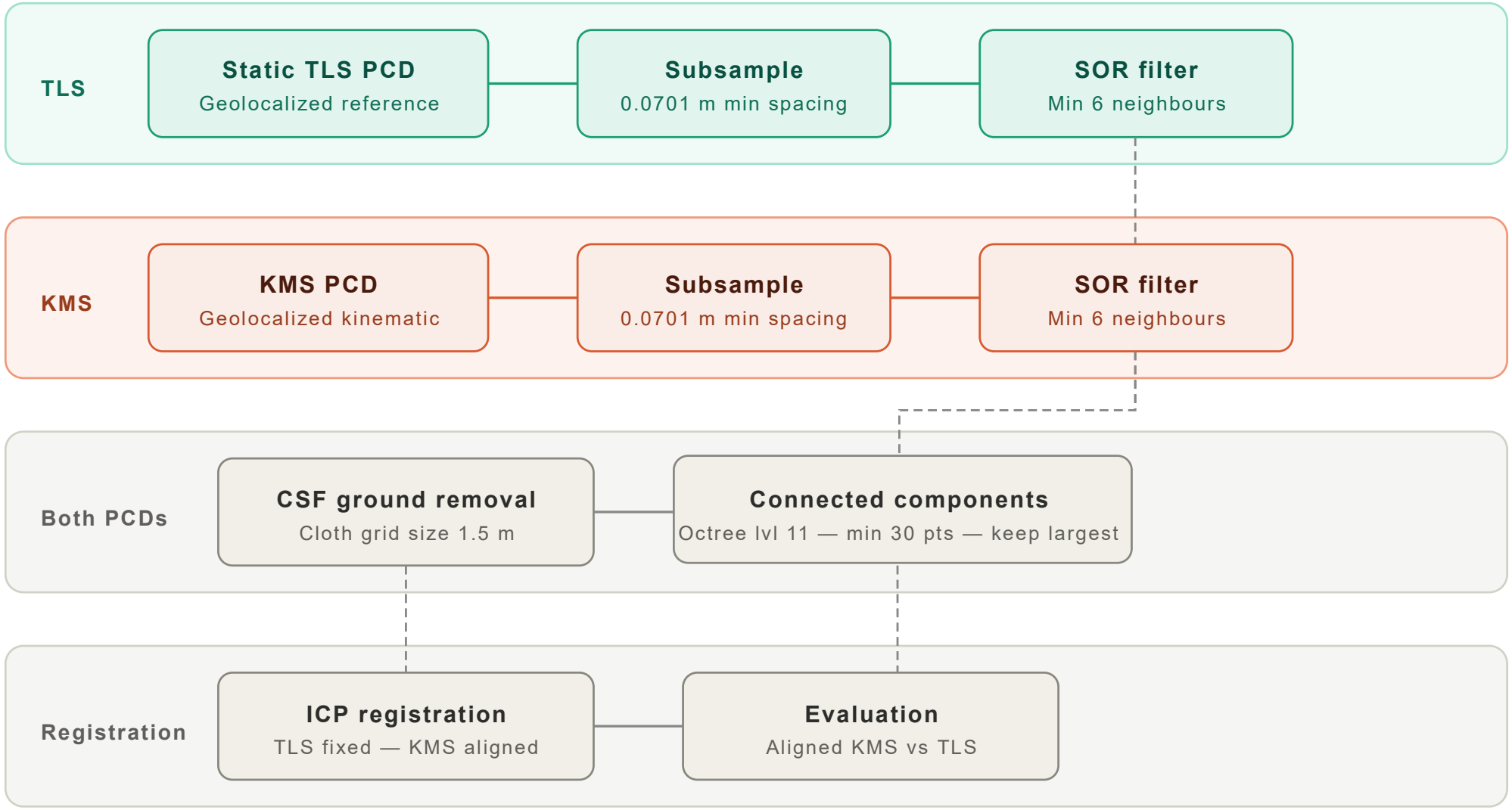
Pointcloud Registration - Kinematic



Georeferenced PCD - Kinematic



Point Cloud Processing



Georeferenced PCD – Static





Georeferenced PCD – Static



Georeferenced PCD - Kinematic



Kinematic Georeferenced Result Overlaps Static Georeferenced Result on OpenStreetMap



Static Georeferenced Result on OpenStreetMap



Kinematic Georeferenced Result Overlaps Static Georeferenced Result on Google Map

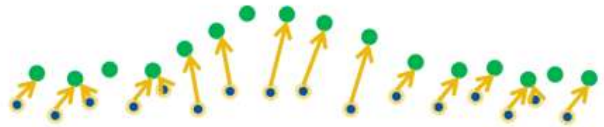
Evaluation

Evaluation Metric

Evaluation Metric

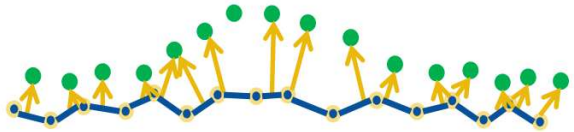
Ⅰ. **Point Cloud Based Comparison**
Cloud-2-Cloud (C2C)

- Nearest distance
- Unsigned values
- Sensitive to density
- Mean, median, p95, std



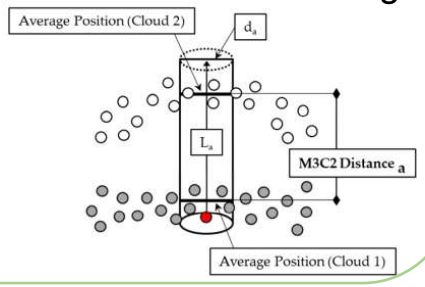
Ⅱ. **Surface Based Comparison**
Cloud-2-Mesh (C2M)

- Normal direction
- Suitable for smooth reference geometry
- Signed values
- Mean, median, p95, std



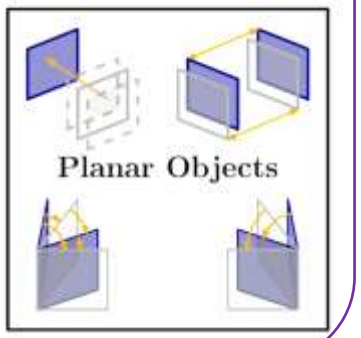
Ⅲ. **Surface Based Comparison**
Multiscale Model-2-Model Cloud Comparison (M3C2)

- Core point, Neighborhood radius D/2, Cylinder of radius d/2 (scale), distance between two average points
- Signed values
- Mean, median, p95, std



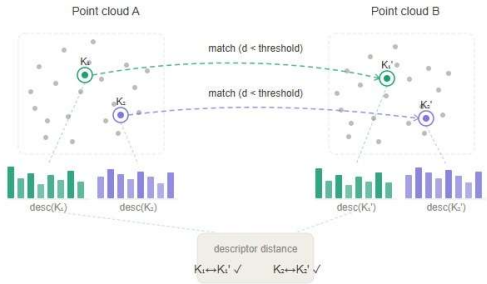
Ⅳ. **Geometry Based Comparison**
Plane fitting

- Extract facades and other planes
- Derive plane parameters
- Compare parameters from multiple passes and scanners
- Compare with Reference



Ⅴ. **Point Based Comparison**
Keypoints

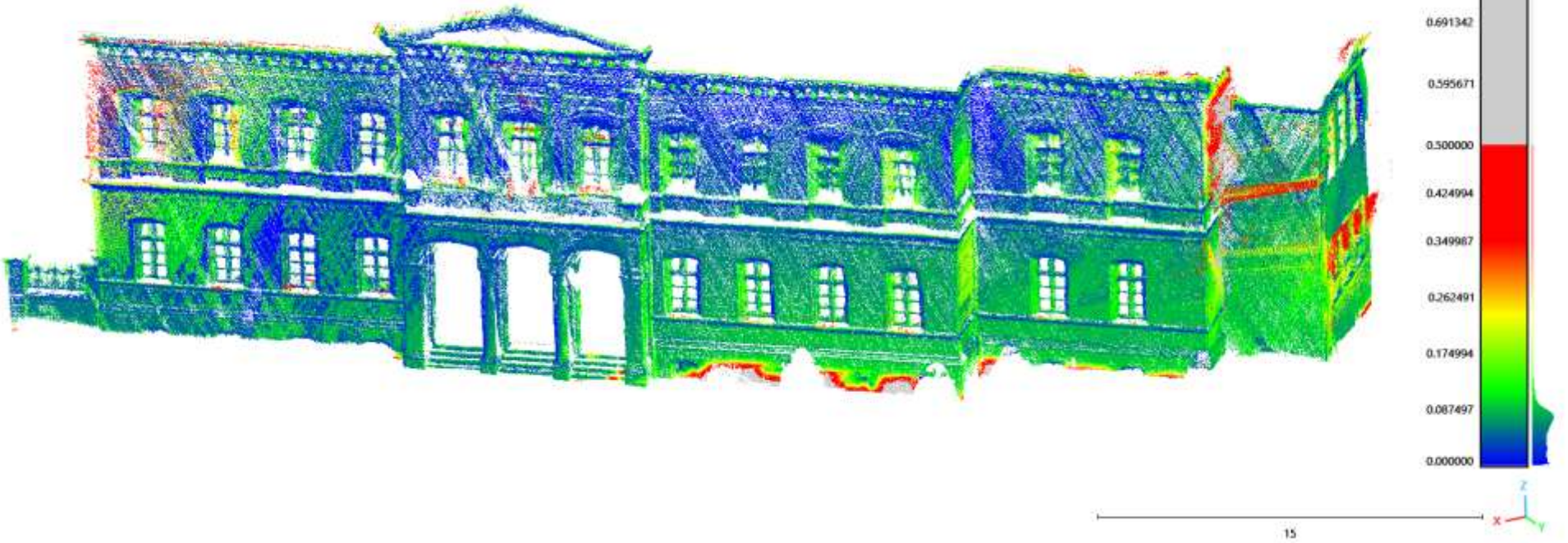
- Pre-process both clouds by downsampling and normal estimation
- Extract stable ISS keypoints from the TLS reference cloud
- Describe and match corresponding points in the KMS cloud with FPFH
- Validate correspondences using normalized L2 distance and a match threshold



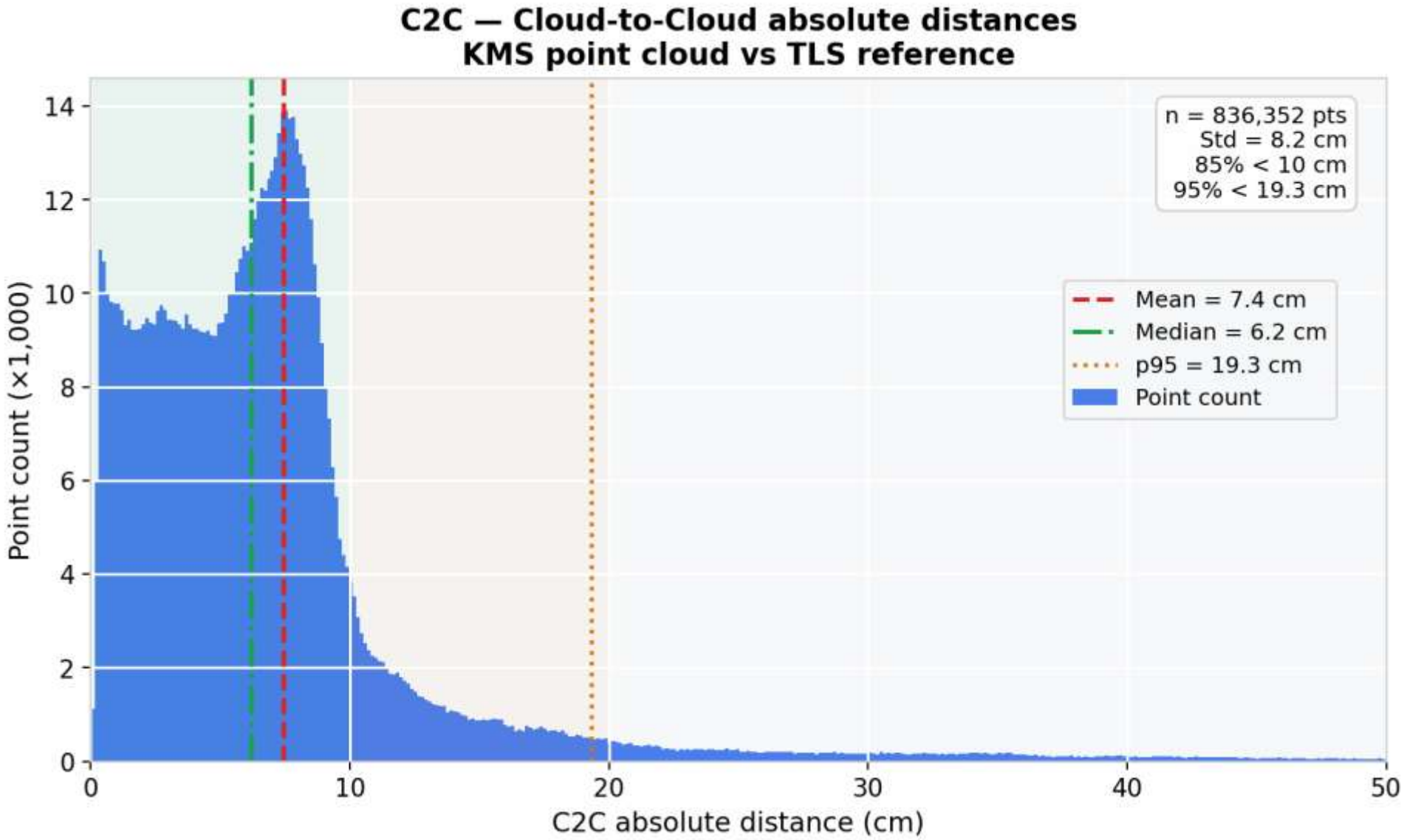
Evaluation- C2C

- Point-to-point distances
- TLS Scanner as reference



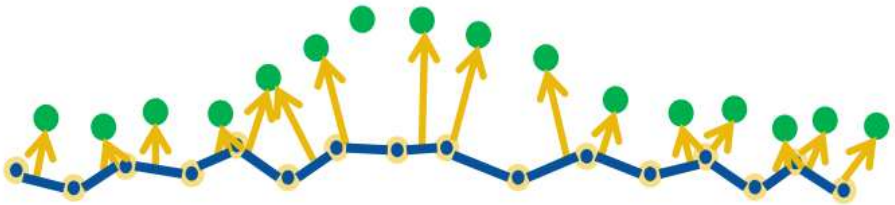


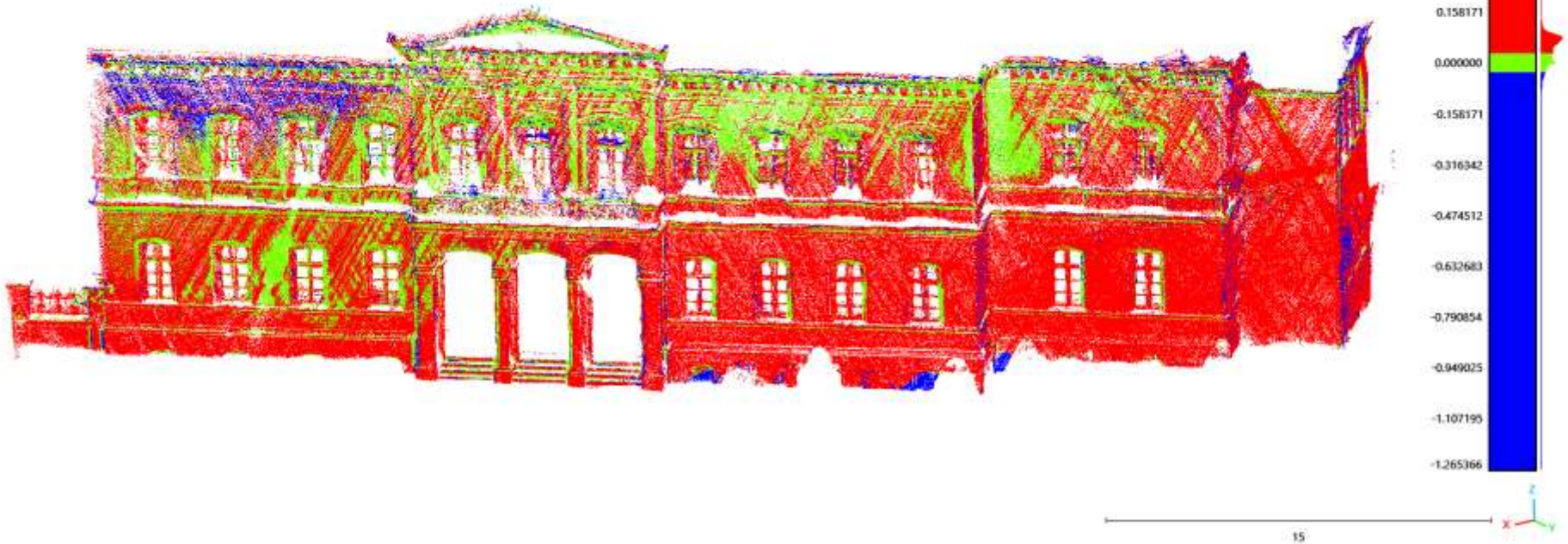
Evaluation- C2C



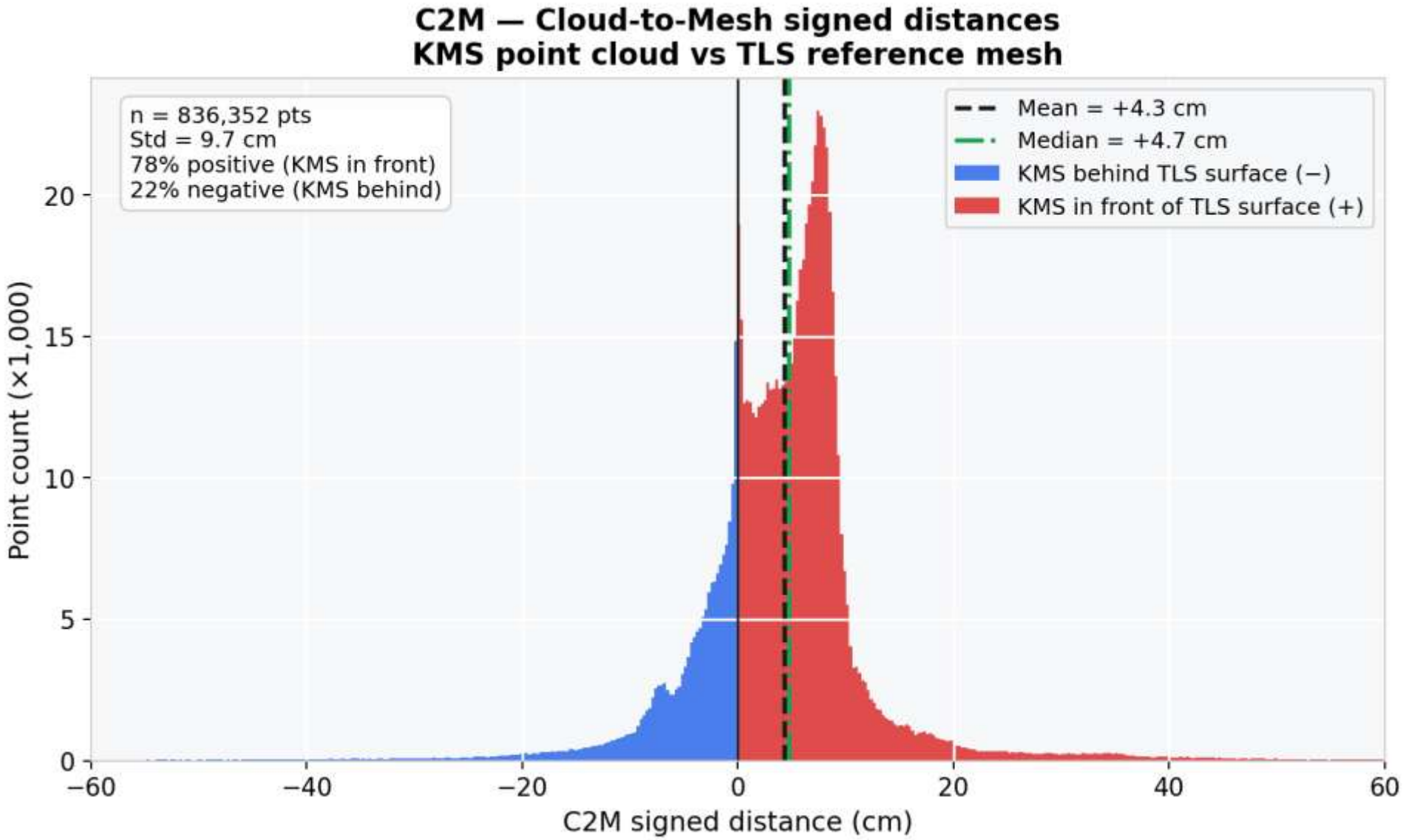
Evaluation- C2M

- Point-to-Mesh distances
- Mesh of TLS reference scan
- Poisson Surface Reconstruction for mesh

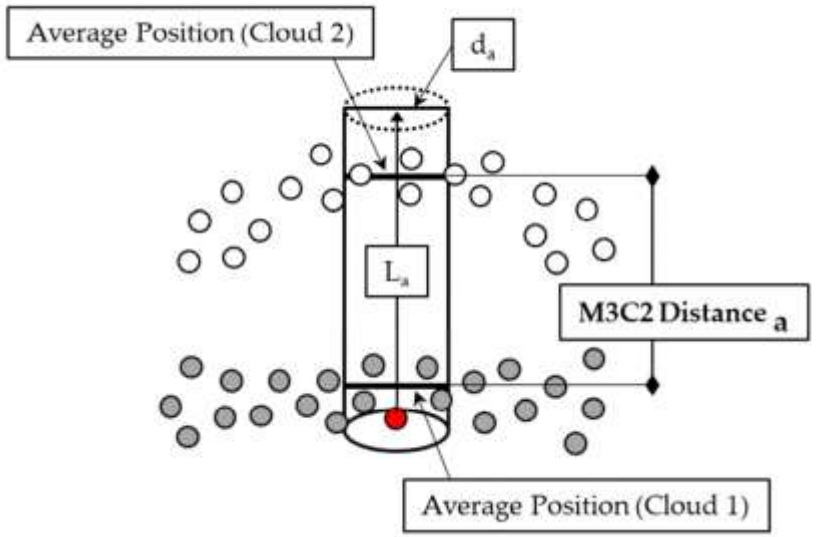


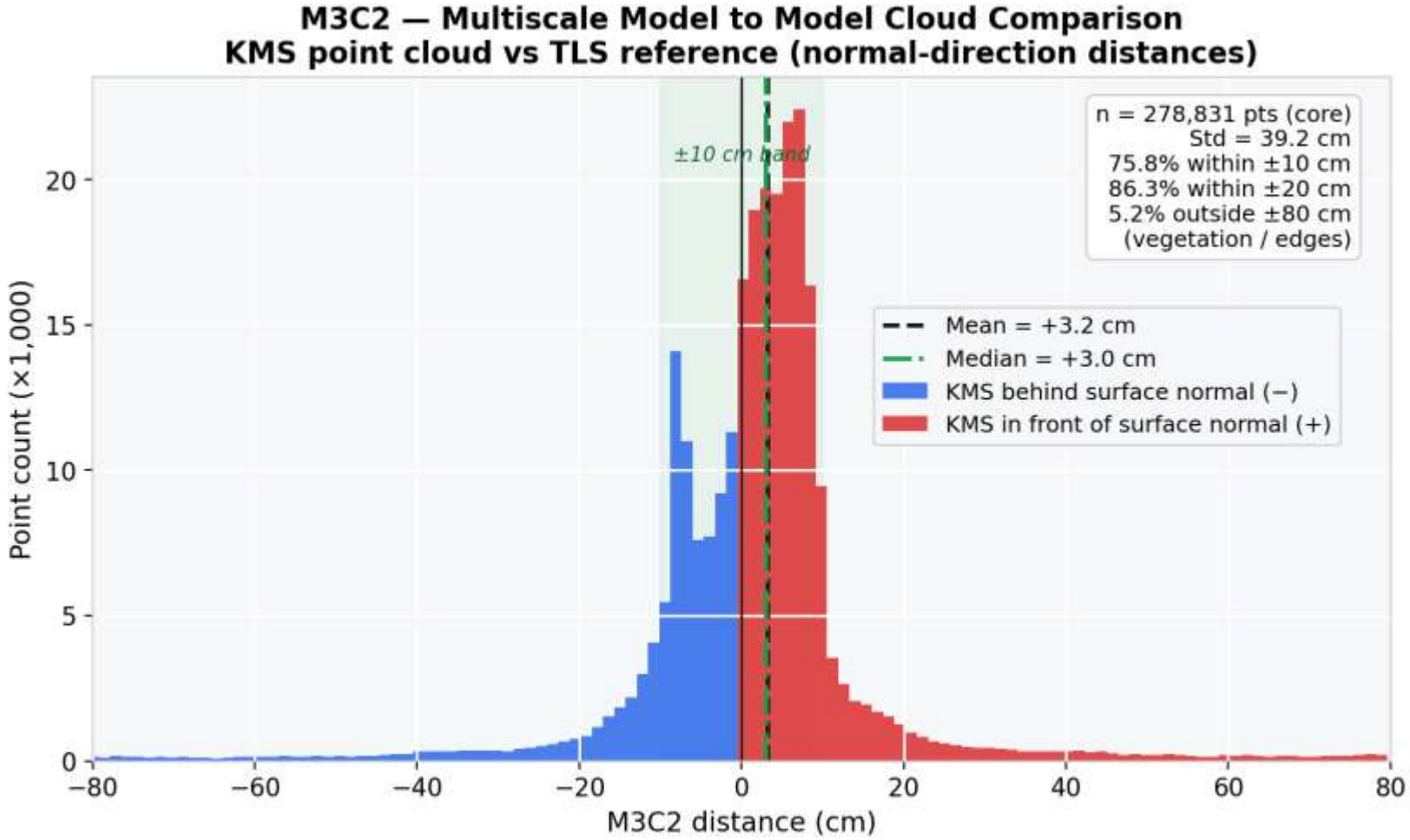


Evaluation- C2M

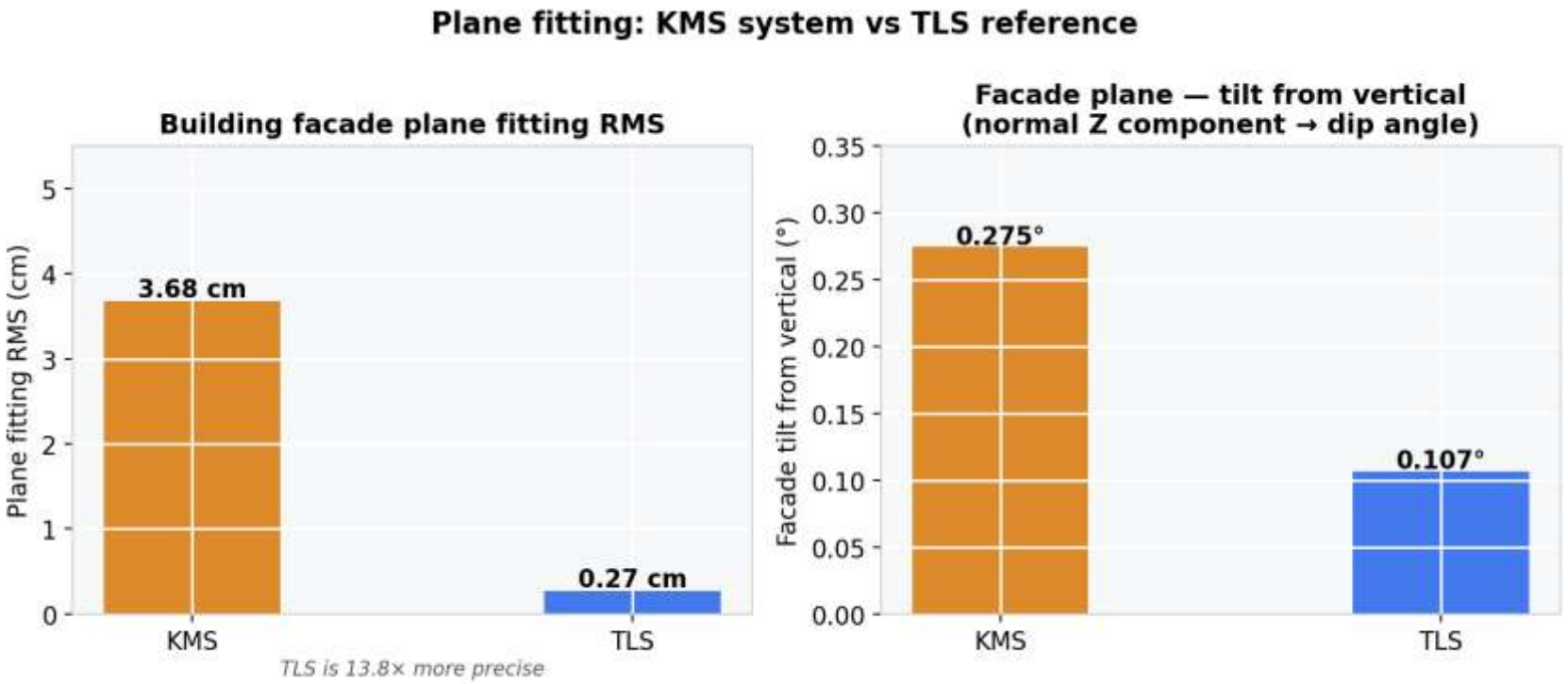


Evaluation- M3C2





Evaluation- Plane fitting

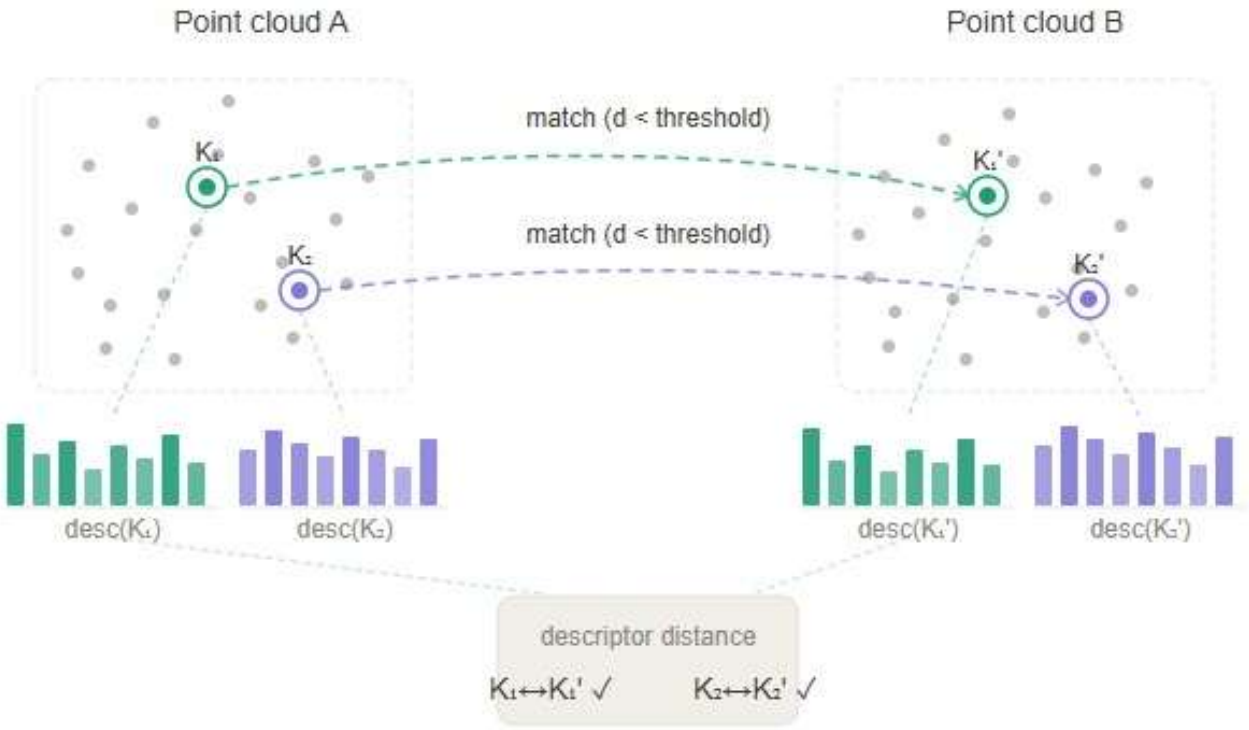




Evaluation- KeyPoints

Pipeline

- Downsample and estimate normals of both point clouds
- Detect ISS key points in TLS cloud
- Select top-k Key points based on stability score
- Find correspondences in KMS cloud
- Compute FPFH descriptors for matched points
- Compute L2 Normalized distance
- Decide match type based on threshold



Evaluation- KeyPoints

- KeyPoints selection: 1500
- Correspondence: 191 Mutual
- Validation: 103 good matches (54%)
  - Thresholds
    - Descriptors: 0.5
    - Spatial distance: 0.05 cm

❖ **Huge density mismatch cab be the reason for Bad type in this case**

REF : 2.7 M

KMS : 0.8 M

Metric	Descriptor Distance	Spatial Distance
Mean	0.4685	0.0308
Minimum	0.2185	0.0019
Maximum	0.8074	0.1547

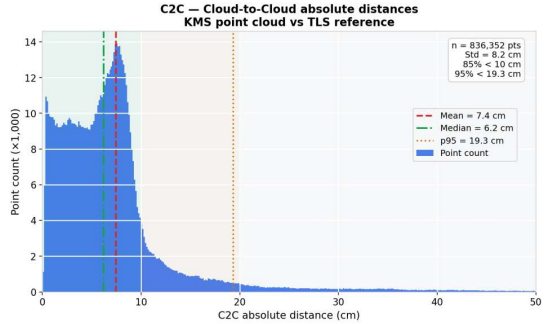
Match Type	Descriptor Distance	Spatial Distance
Good	0.218	0.011
Good	0.231	0.011
Good	0.249	0.007
Bad	0.788	0.016
Bad	0.744	0.011
Bad	0.654	0.002

Summary of findings

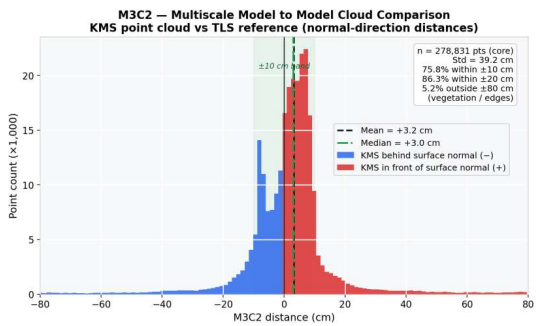
KMS is effective for rapid scene capture, but TLS remains more stable geometric reference

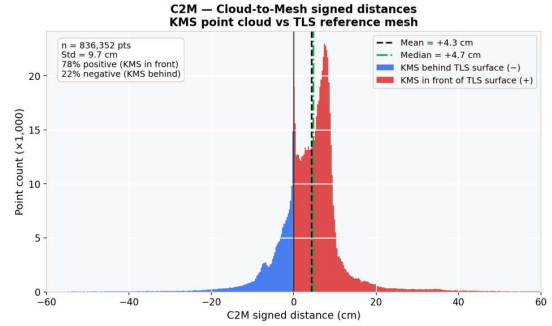
- **Overall alignment.**
- **Systematic signed bias**
- **Lower local surface stability**
- **Structural similarity, not equivalence (point-based result)**

C2C: overall distance level

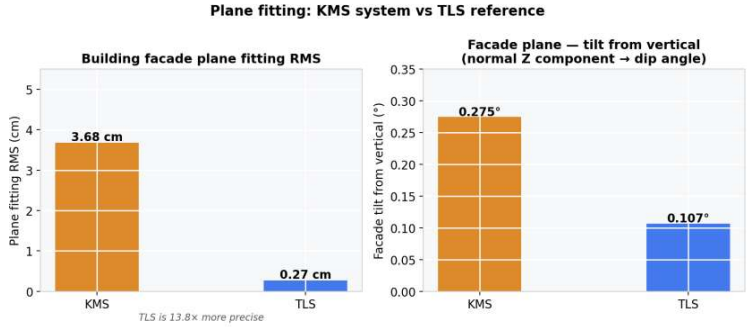


C2M and M3C2: signed offset tendency





Plane fitting: facade quality gap



Conclusion

KMS reproduces the overall building geometry well, but TLS remains the high-precision reference for evaluation.

KMS is suitable for

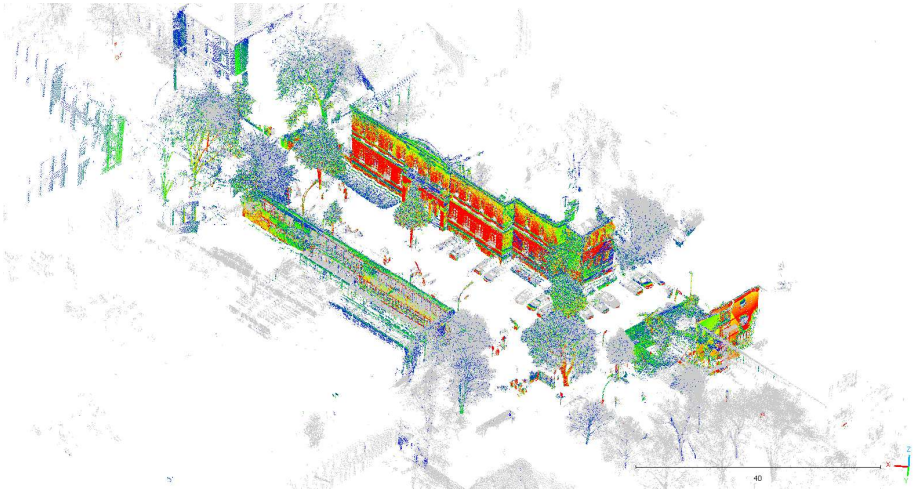
- efficient large-area 3D acquisition
- overall geometry reconstruction and facade-position checks
- rapid follow-up surveys where centimeter accuracy is acceptable

TLS remains needed for

- precise planar analysis and reference modelling
- local detail, edges and stable surface geometry
- evaluation of mobile mapping results



TLS Georeferenced Pointcloud

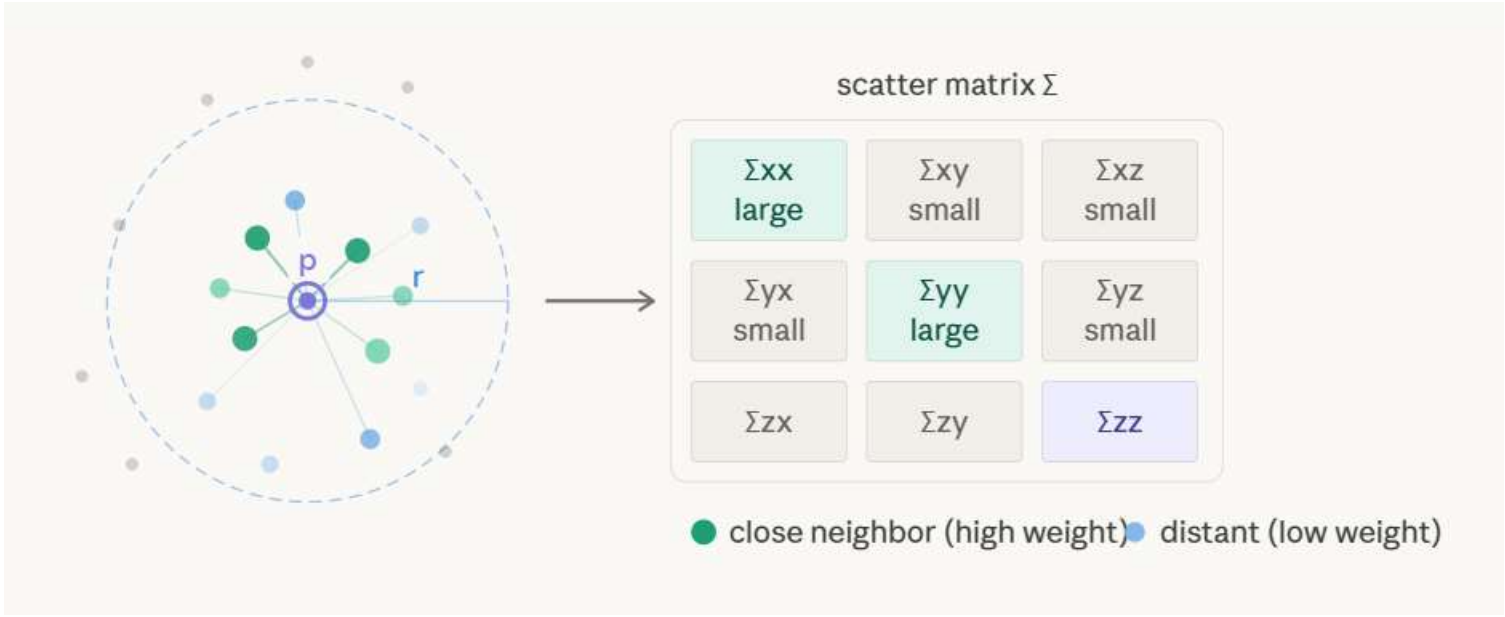


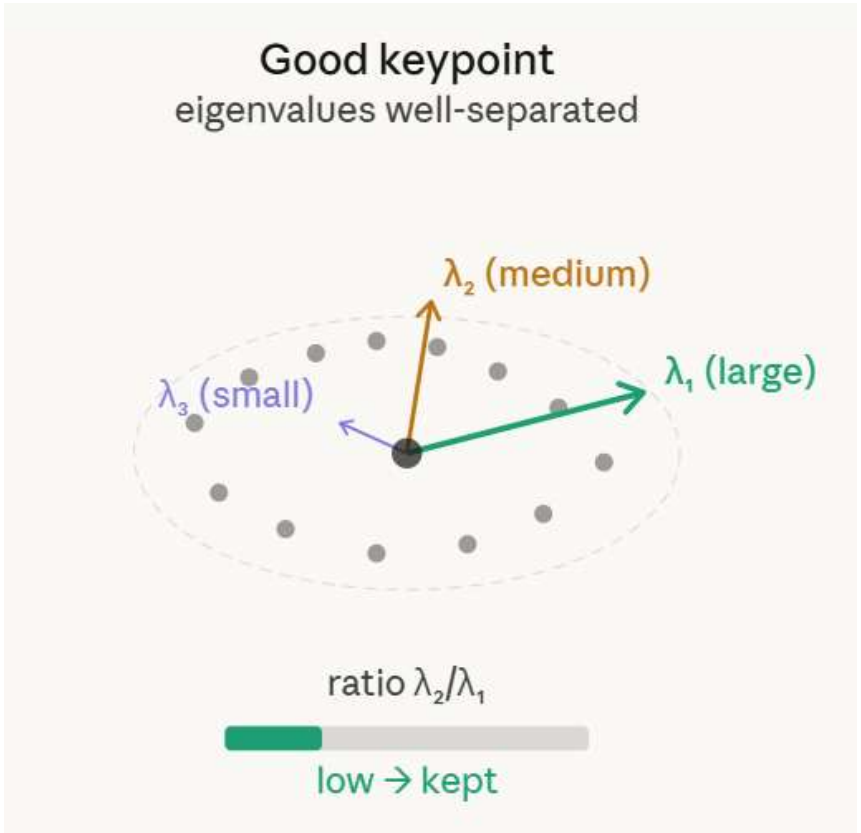
KMS Georeferenced Pointcloud

Thank you for your attention!

Evaluation- KeyPoints

ISS KeyPoints Detector





Evaluation- KeyPoints

UNIVERSITÄT BONN IGG

Stability Score

$$\text{score} = \frac{\lambda_3}{\lambda_1} \cdot \left(1 - \frac{\lambda_2}{\lambda_1}\right)$$

- λ_3 / λ_1 **large** → strong 3D variation
- λ_2 / λ_1 **small** → not flat

So high score = good keypoint

Geometry	Eigenvalues
Flat surface	$\lambda_1 \approx \lambda_2 >> \lambda_3$
Edge	$\lambda_1 >> \lambda_2 \approx \lambda_3$
Corner	$\lambda_1 \approx \lambda_2 \approx \lambda_3$